HSLU Lucerne University of Applied Sciences and Arts

Chemical bonds

CHEM: Chemistry

Dr. Rebecca Ravotti
HS25

Engineering and Architecture
October 7, 2026

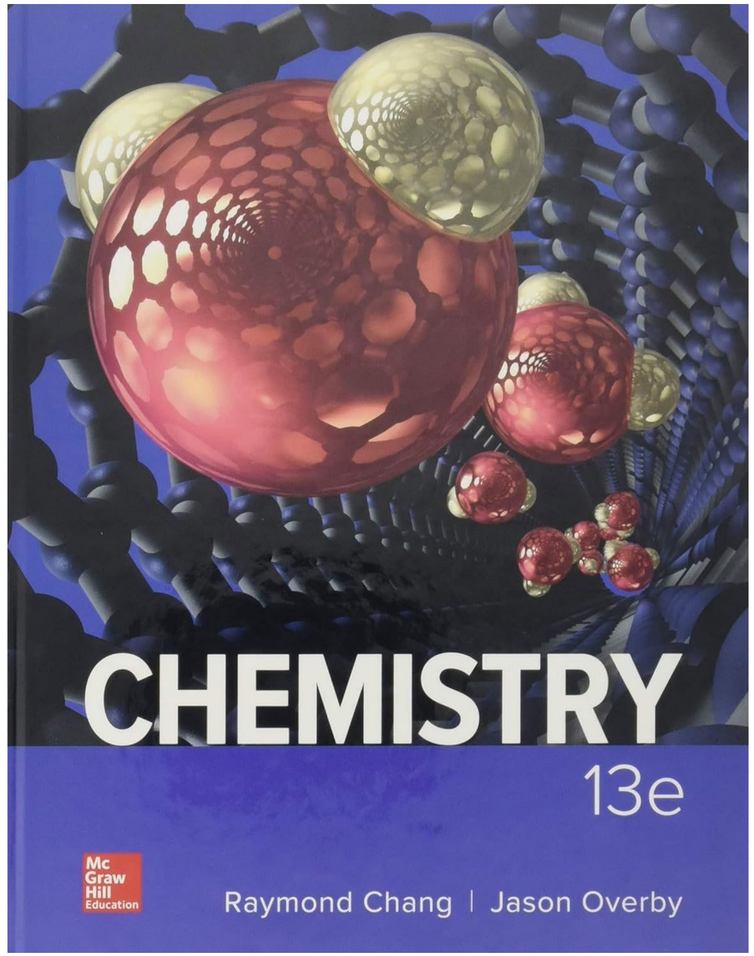
Course Details

Lecturer	Date	Week	Topic
Macarena San Martin	16.09.2026	1	Introduction to Matter
Macarena San Martin	23.09.2026	2	Atomic Structure and Radioactivity
Macarena San Martin	30.09.2026	3	Atomic Models (Self Study), Periodic table and electronic configuration
Rebecca Ravotti	07.10.2026	4	Chemical Bonds and Hybridization
Rebecca Ravotti	14.10.2026	5	Lewis Structure
Davide Toniolo	21.10.2026	6	Chemical Reactions
Davide Toniolo	28.10.2026	7	Acids & Bases
Davide Toniolo	04.11.2026	8	Acids & Bases Exercises
Davide Toniolo	11.11.2026	9	Redox Reactions
Davide Toniolo	18.11.2026	10	Redox exercises
Rebecca Ravotti	25.11.2026	11	Metals and Semimetals
Rebecca Ravotti	02.12.2026	12	Organic Compounds nomenclature
Rebecca Ravotti	09.12.2026	13	Organic Exercises
Macarena /Rebecca/Davide	16.12.2026	14	Final Laboratory

Recommended literature

Chemistry ; Authors, Raymond Chang, Jason Overby ; Edition, **13, illustrated** ; Publisher, McGraw-Hill Education, 2018 ; ISBN, 1260085317, 9781260085310

Periodic table: **https://ptable.com**



Structure of the Lecture

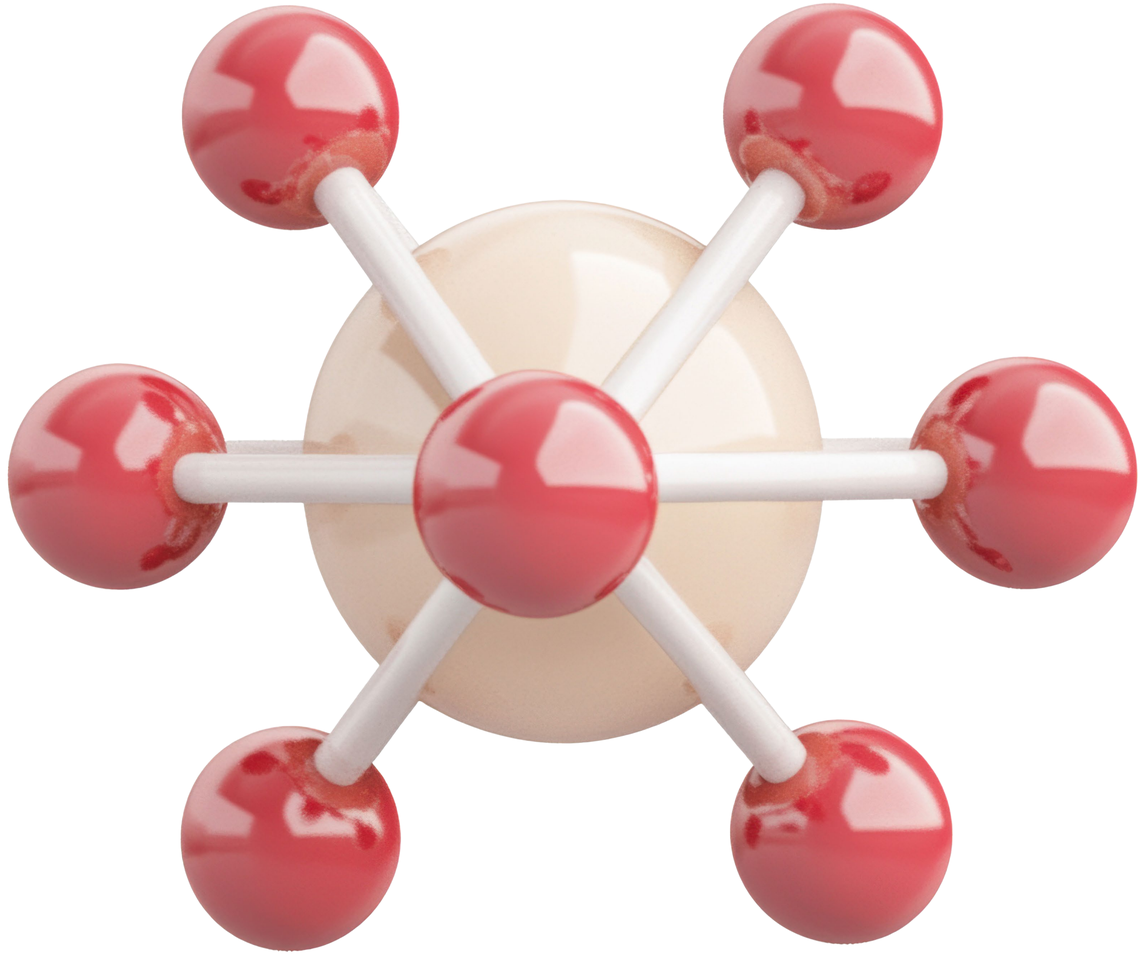
- Introduction/learning objectives (5 min) (08:30-08:35)
- Recap orbitals (10 min) (08:35-08:45)
- Theoretical input (30 min) (08:45 – 09:15)

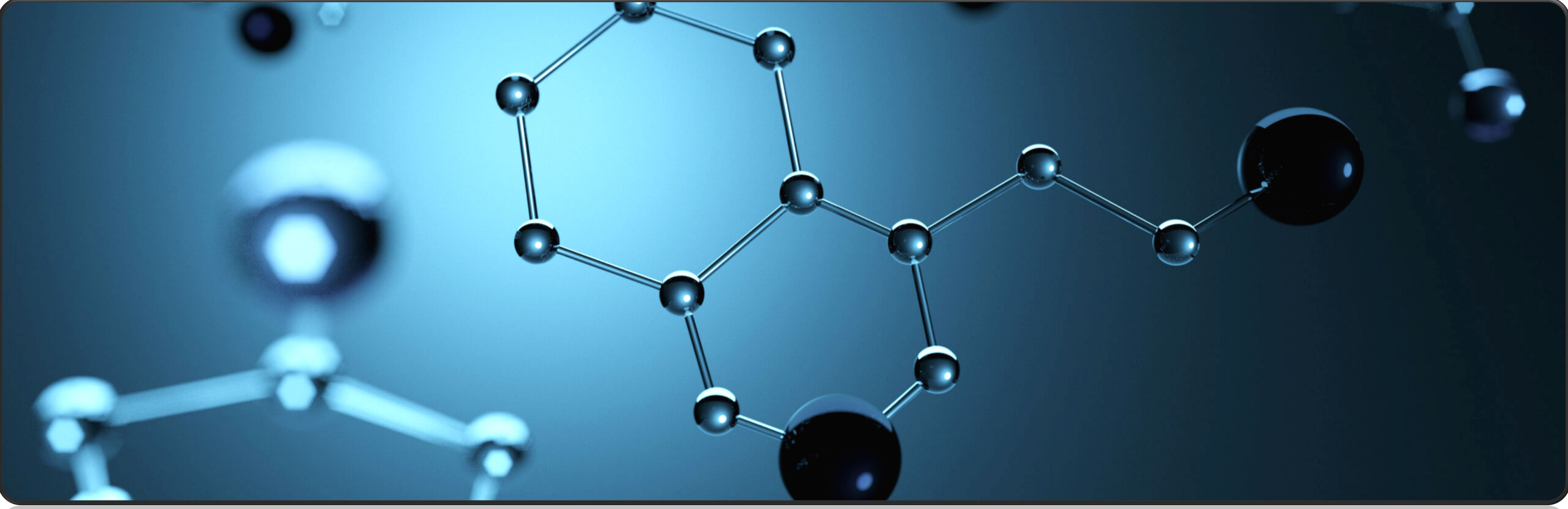
Break (20 min) (09:15 – 09:35)

- Theoretical input (30 min) (09:35 – 10:05)

Break (20 min) (10:05 – 10:25)

- Small theoretical input (10 min) (10:25 – 10:35)
- Case study + Kahoot (15 min) (10:35 – 10:50)
- Wrap-up/Feedback (5 min) (10:50 – 10:55)





Today's Content

- Octet rule
- Hybridization
- Types of bonds
- Concept of polarity
- Correlation of bonds to physical properties (e.g. conductivity, melting points)

Learning objectives

Know the types of bonds

Be able to distinguish between bonds

Be able to understand the theory between different bonds

Be able to correlate bonds to physical properties

Theory of bonds

What are chemical bonds?

"an attraction between two or more atoms that form a chemical"

- Based off exchanging/sharing <u>valence</u> electrons
- Can be found <u>everywhere</u>!
- Different types of different strengths exist
- Influence properties (high/low melting points, solubility, electrical conductivity, etc.)
- Follow the **<u>octet rule</u>**





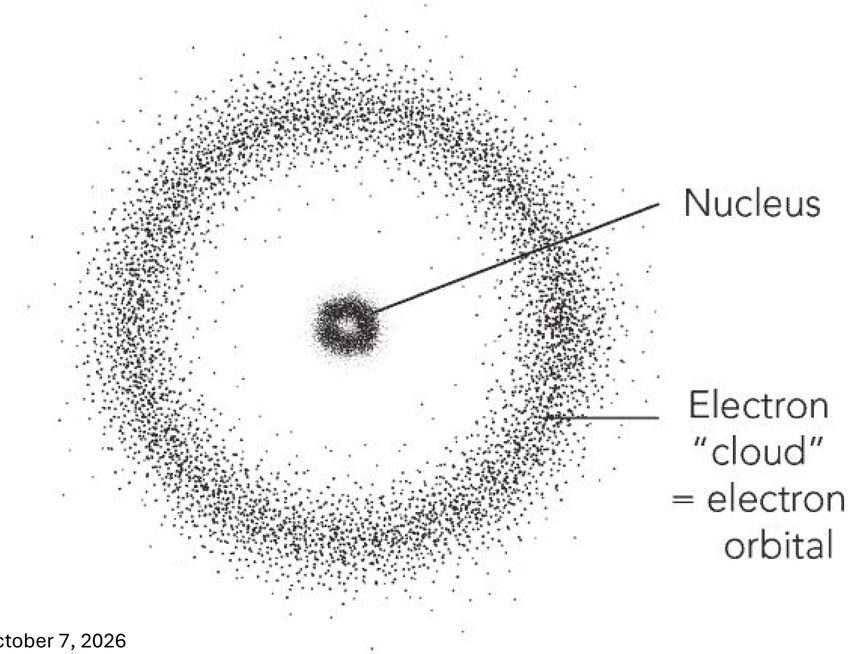


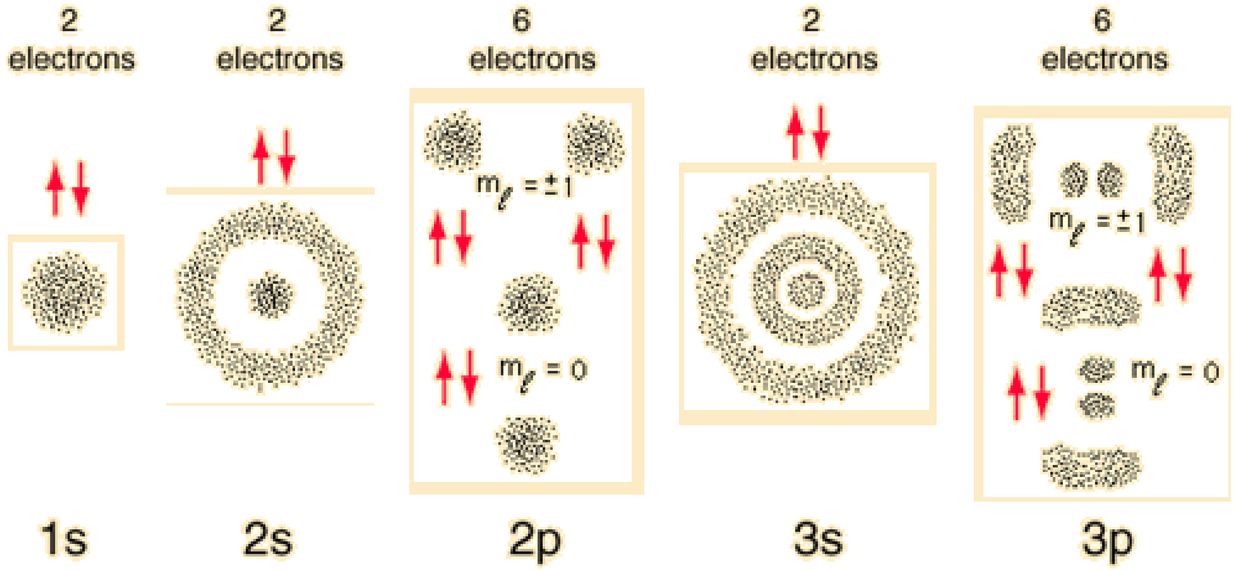


Recap - Orbitals

- We cannot know both location and speed of an electron at the same time!
  →Either one or the other

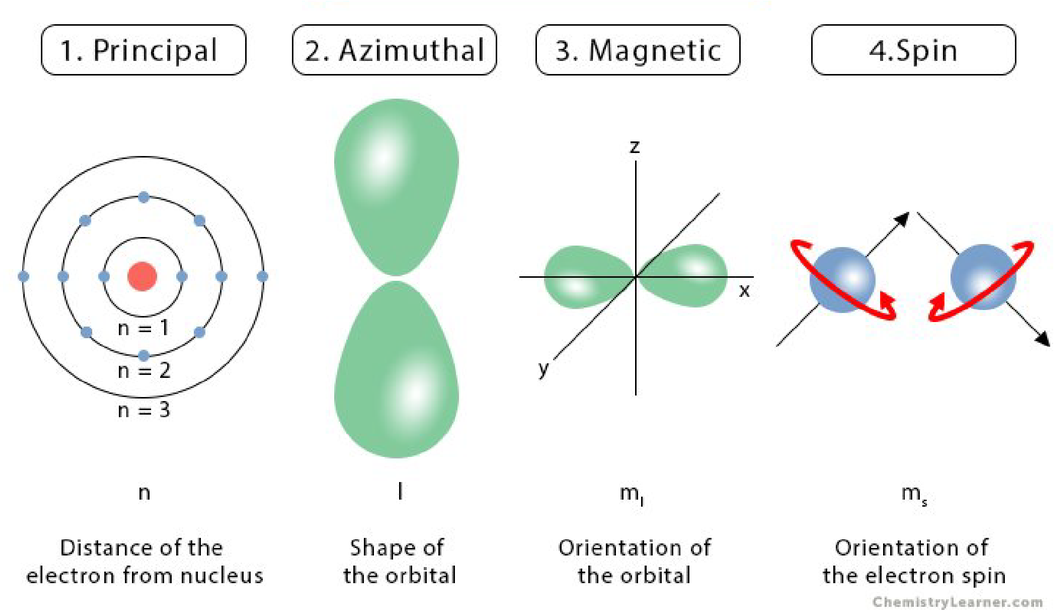
- **<u>Orbitals</u>** → describe the probability of finding an e- around the nucleus! Calculated with wave functions

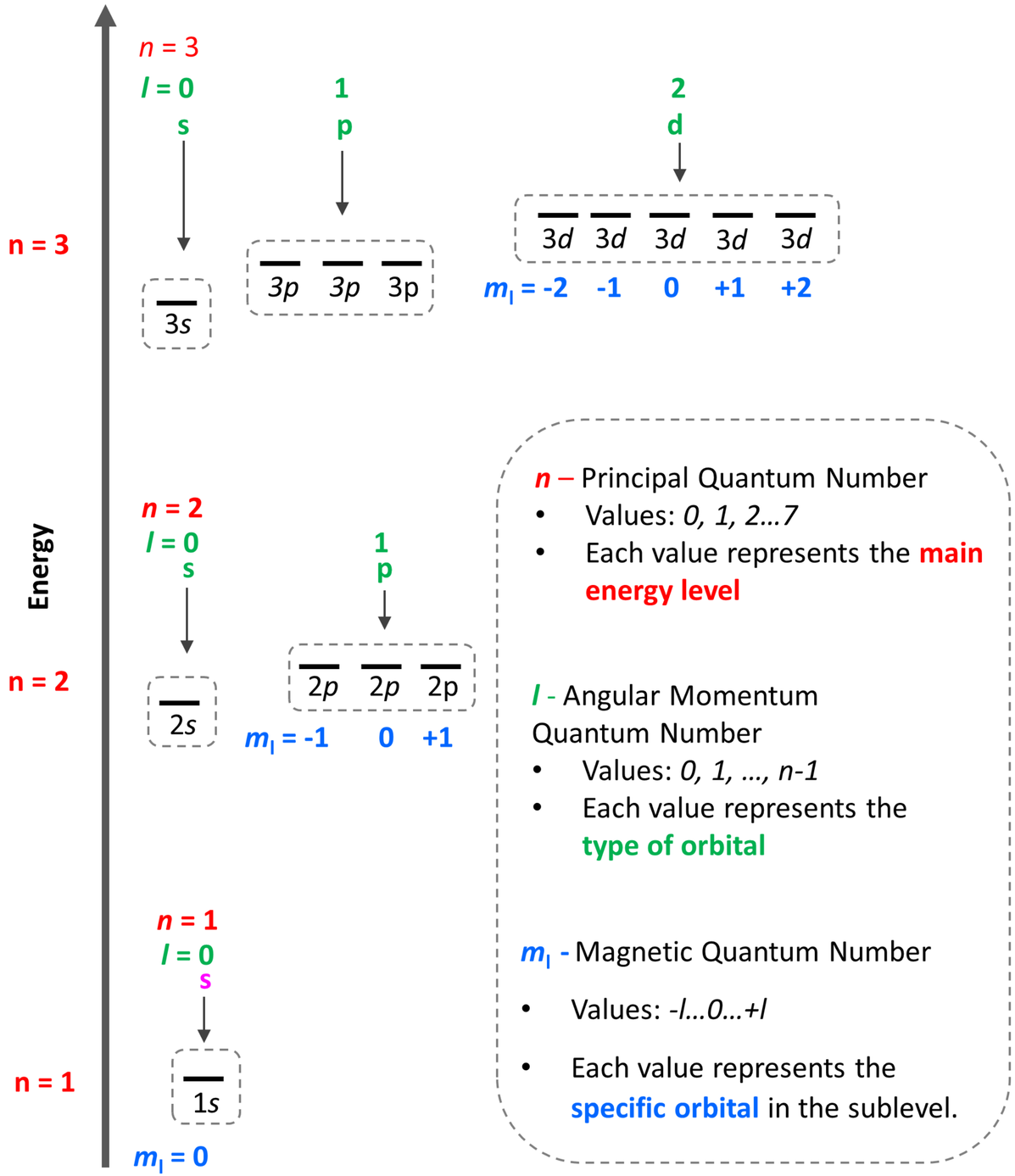




Recap – Quantum numbers

Similar to an ID or home address description for electrons!





n – Principal Quantum Number

- Values: *0, 1, 2…7*
- Each value represents the **main energy level**

l - Angular Momentum Quantum Number

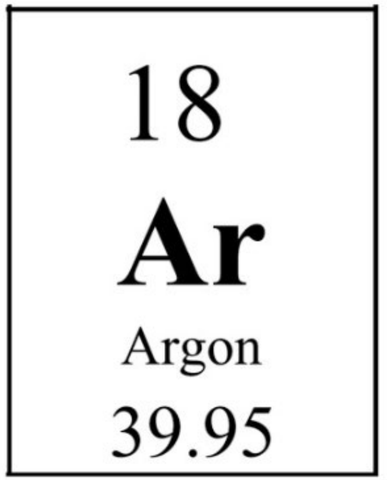
- Values: *0, 1, …, n-1*
- Each value represents the **type of orbital**

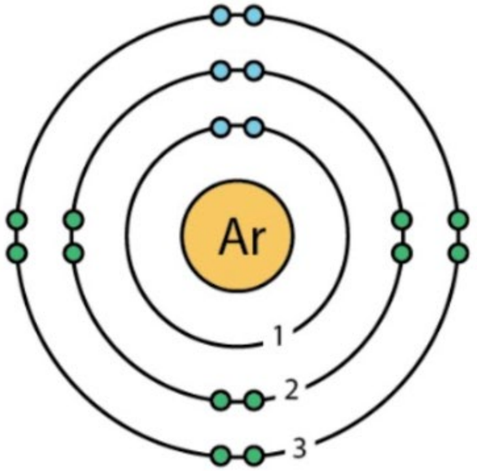
m_l - Magnetic Quantum Number

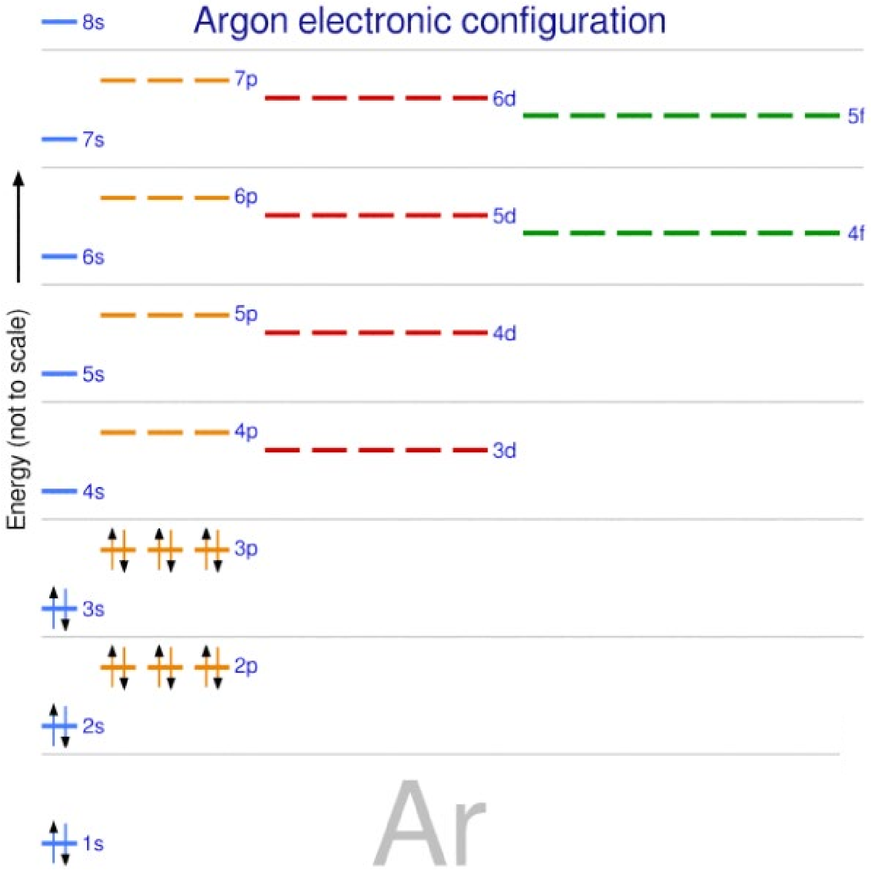
- Values: *-l…0…+l*
- Each value represents the **specific orbital** in the sublevel.

Octet rule

Noble gases are super stable! Why?







More in-depth explanation for nerds:

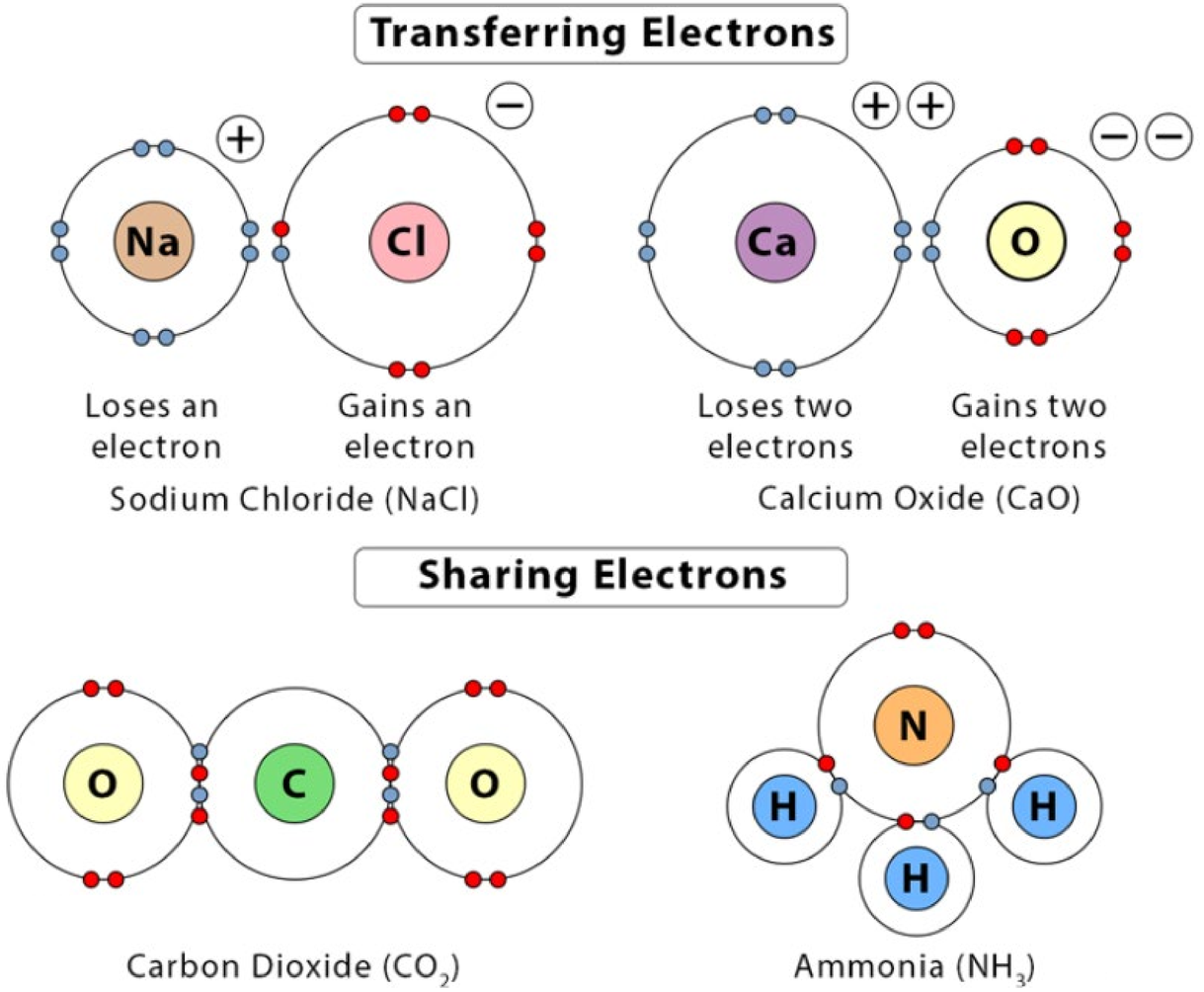
https://www.youtube.com/watch?v=KX1323EmWwg

Octet rule

All elements want to be noble

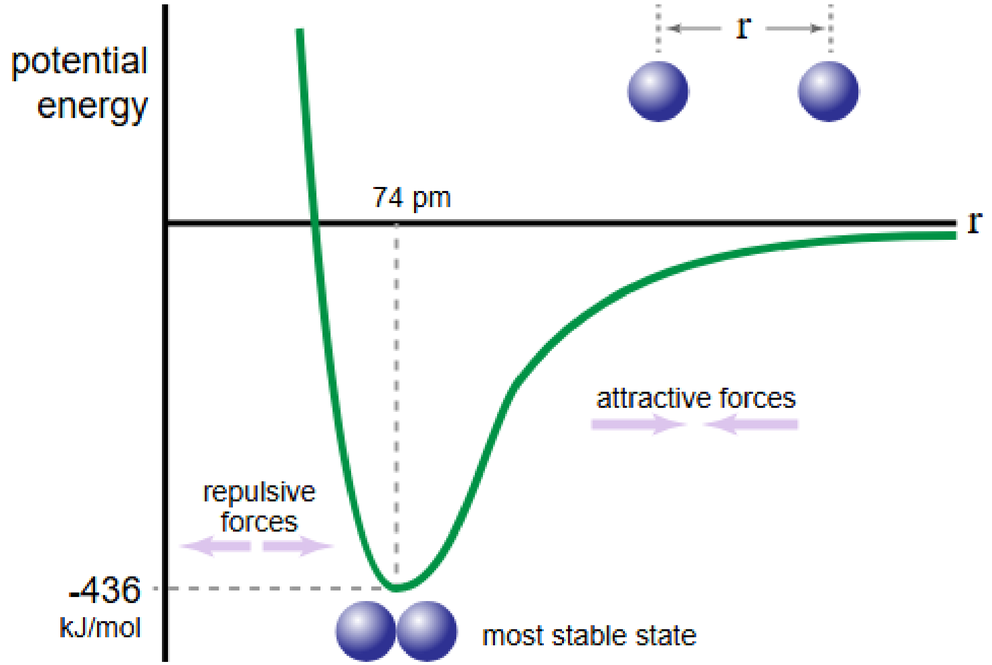
Noble gases watching other elements trying to achieve fully-filled orbitals:





Why do atoms bond?

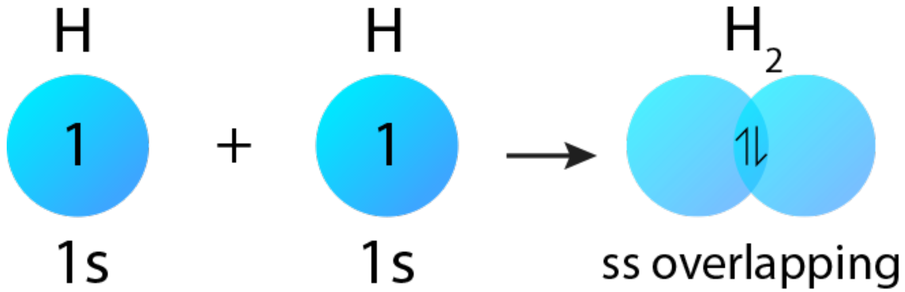
- To achieve the most stable configuration they can
- Atoms have potential energy due to attractive/repulsive forces
- When the distance between two atoms is “optimal”, there’s a potential energy minimum → bond!



How do atoms bond? Overlap

- Orbitals where valence electrons are found overlap

- Depending on the type of orbitals overlapping, different bonds are formed

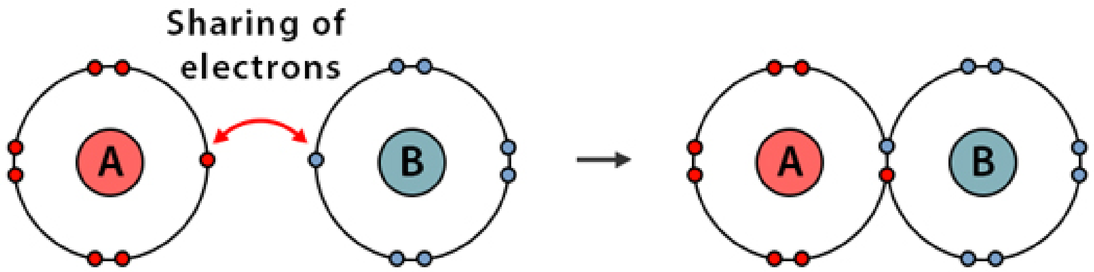


- Some valence electrons don't participate in bonds, but stay around the atom as "free electrons" → usually in pairs → "free" electron pairs

Covalent bond

The e^- are shared! → each atom reaches the octet

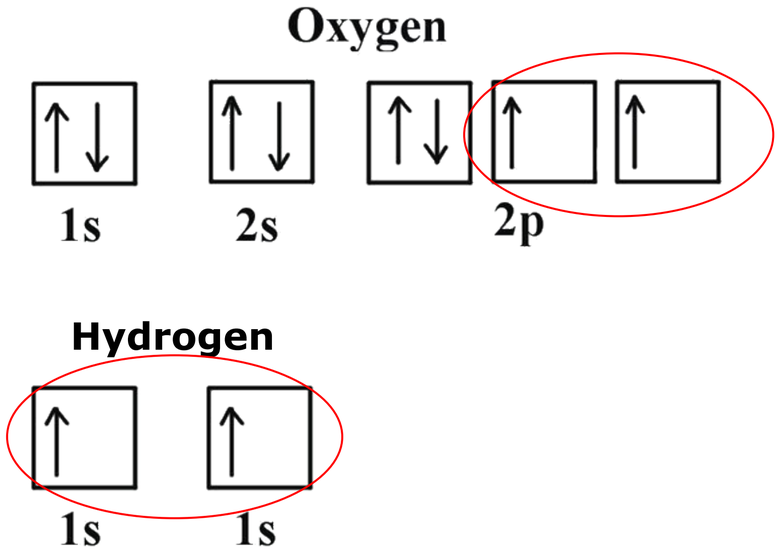
→**<u>Covalent</u>** bond

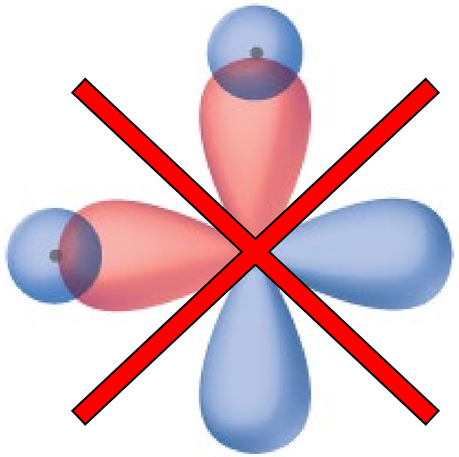


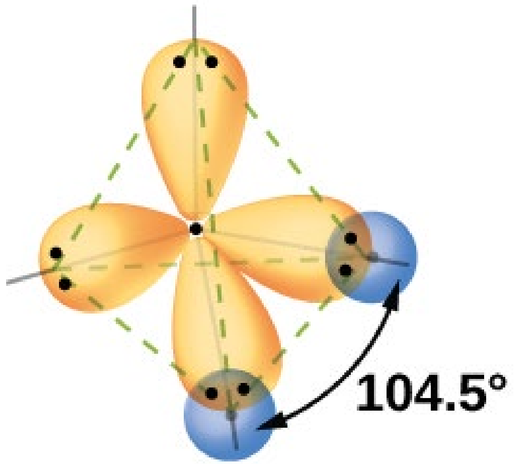
Different types of covalent bond form depending on hybrid/non-hybrid orbitals involved

The strange case of H_2O

- We should expect H_2O to have 90° angles
- Experimental evidence shows it's not the case → why?
- Reality is a bit more complex!





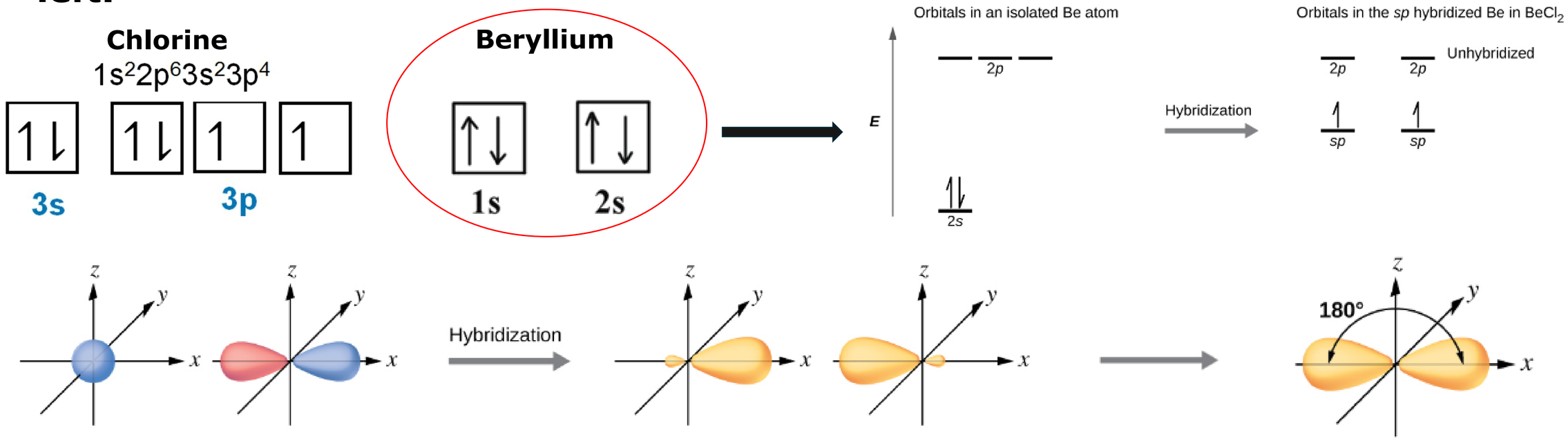


How do atoms bond? Hybridization - sp

- Atoms try to "prepare" for bonding by optimizing their energy/shape

  → orbitals of equivalent energies can "merge" → make new hybrid orbitals!

- Example: $BeCl_2$ → **sp hybrid orbitals! (1s + 1p orbitals) → two "empty" p orbitals left!**



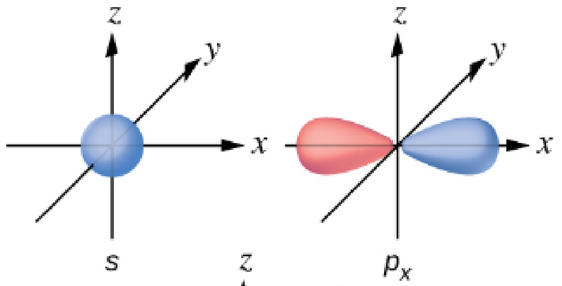
- Hybridization happens **within the atom**!

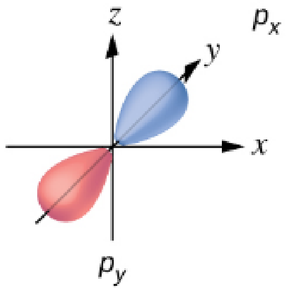
Hybridization – sp^2

- 1 s + 2 p orbitals

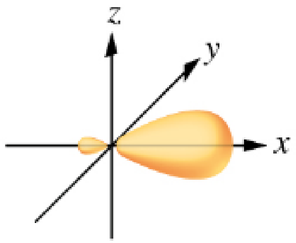
= **3 sp^2 orbitals! → 1 p left!**

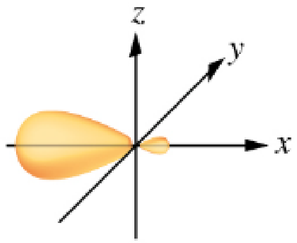
- Leads to a 120° configuration

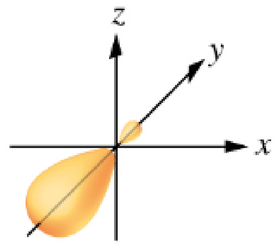




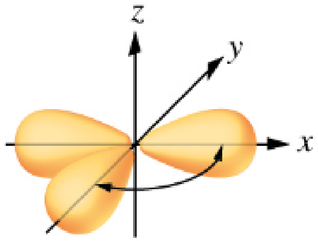
Hybridization





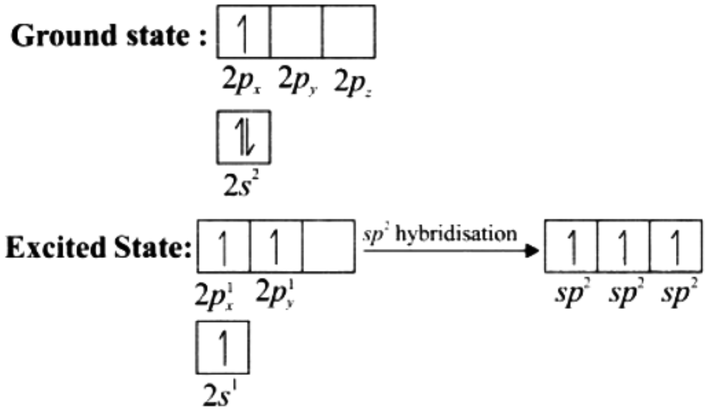


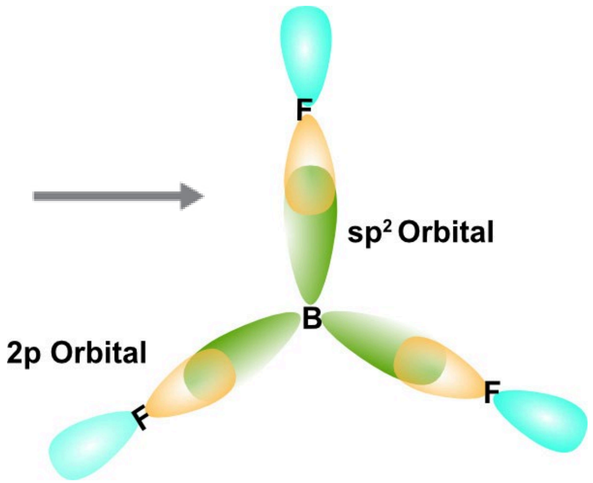
Gives a trigonal planar arrangement



- Example: BF_3

Boron





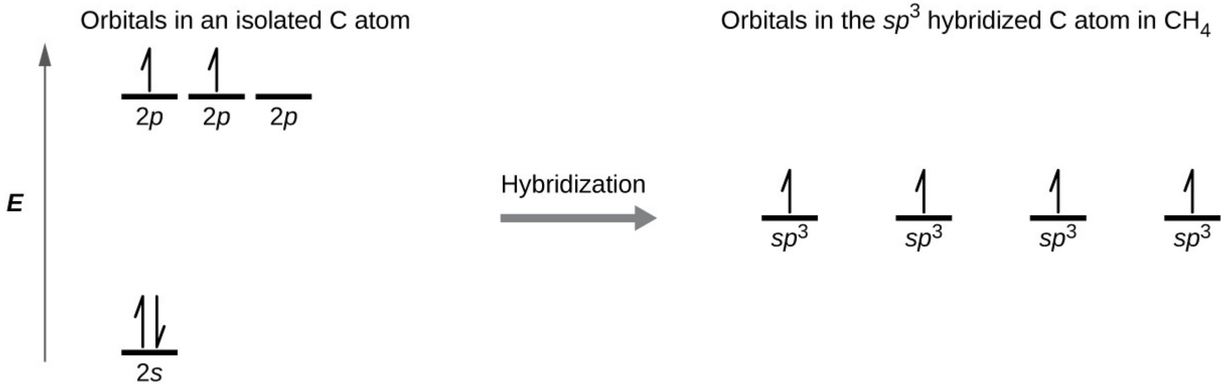
Hybridization – sp^3

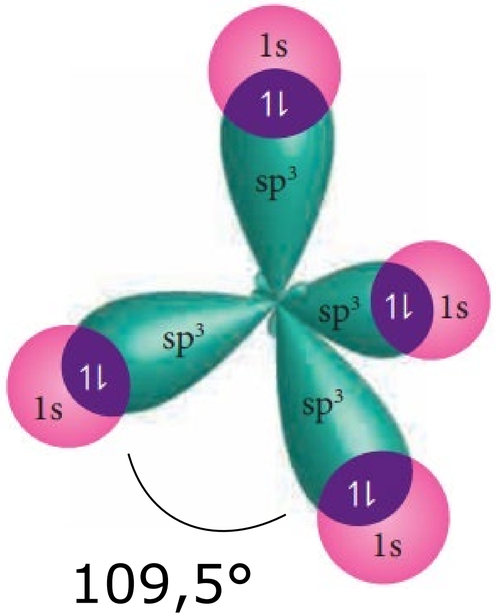
- 1 s + 3 p orbitals

= **4 sp^3 orbitals! – no p left!**

- Leads to a 109,5° configuration

- Example: CH_4

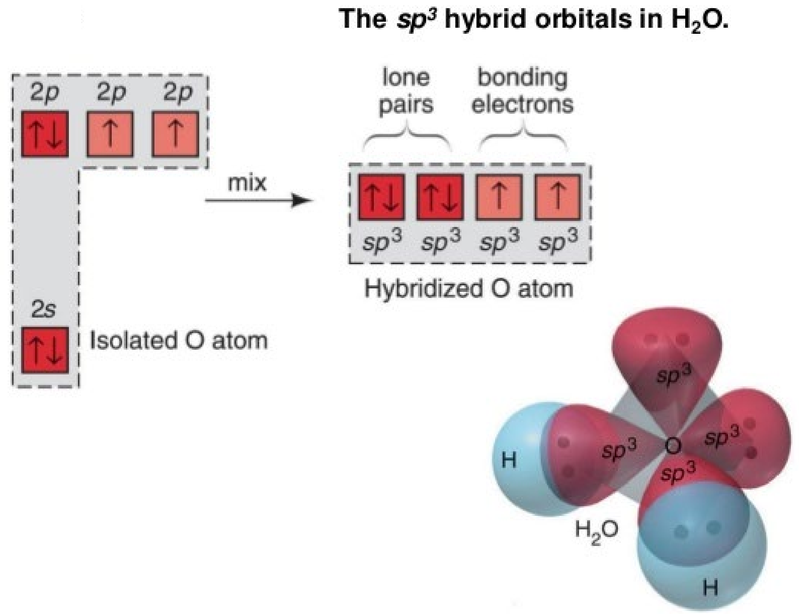


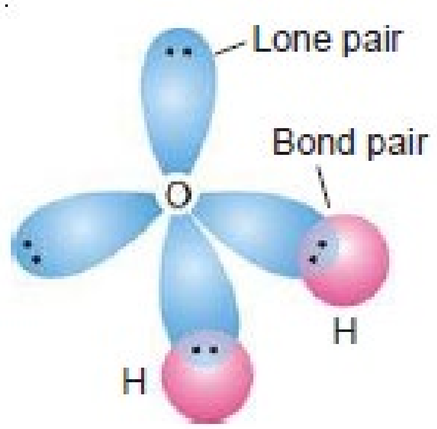


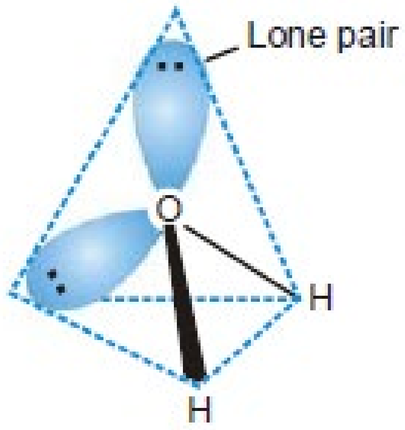
Hybridization – sp^3 – H_2O vs CH_4

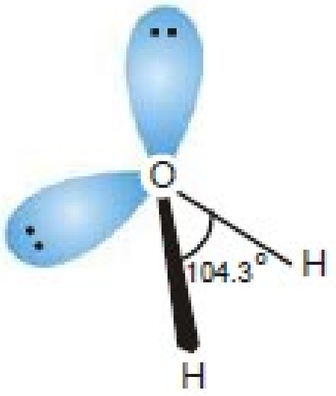
- H_2O also hybridizes to sp^3
- Should have a 109,5° configuration

- Ends up with 104° angle because of lone pair repulsion









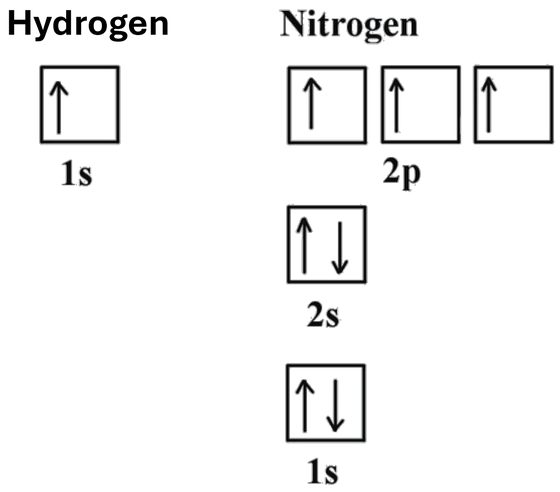
Hybridization – key points

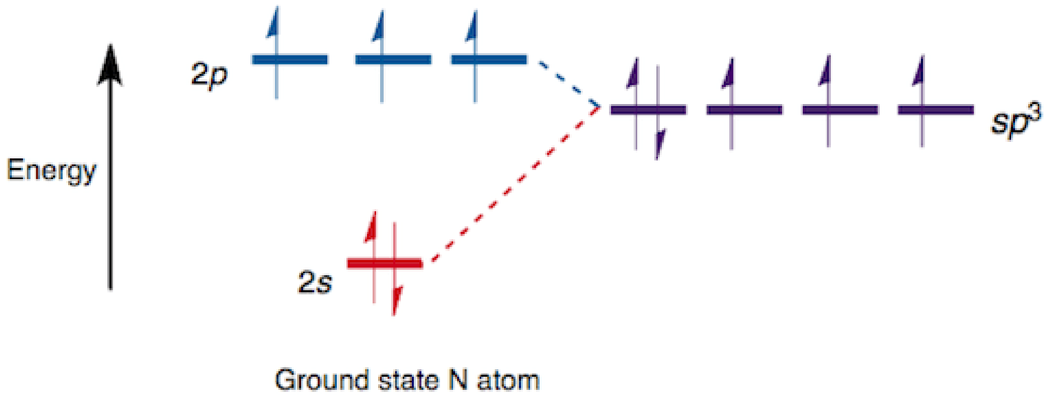
- Atomic orbitals with similar energies undergo hybridization
- The number of hybrid orbitals formed is equal to the number of atomic orbitals mixing
- Not only half-filled orbitals participate, even completely filled orbitals can participate
- The shape of the molecule can be predicted
- **Hybridization happens only during the bond formation!**

coordination number of the carbon atom	hybridization	geometry	# orbitals participating in hybridization
4	sp^3	tetrahedral	4 (1s + 3p)
3	sp^2	planar trigonal	3 (1s + 2p)
2	sp	linear	2 (1s + 1p)

Exercise

Determine the hybridization of NH_3



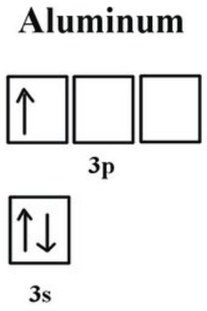


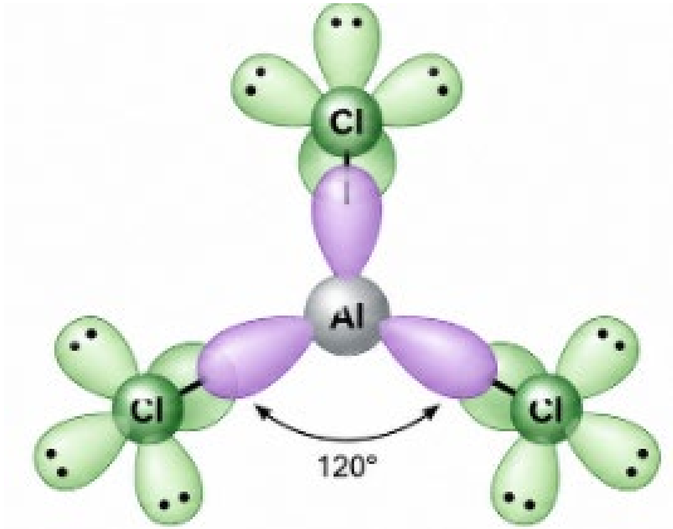
Break
09:15 – 09:35

Types of bonds (intramolecular)

Exercise

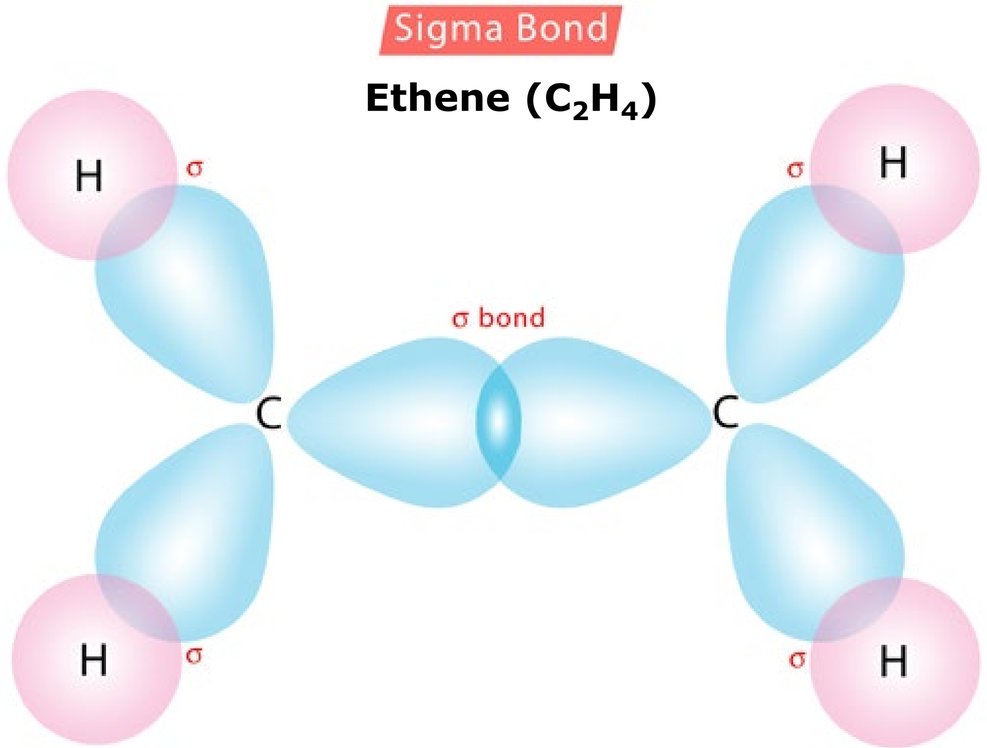
Determine the hybridization of $AlCl_3$ (central atom)





Covalent bond – σ bond

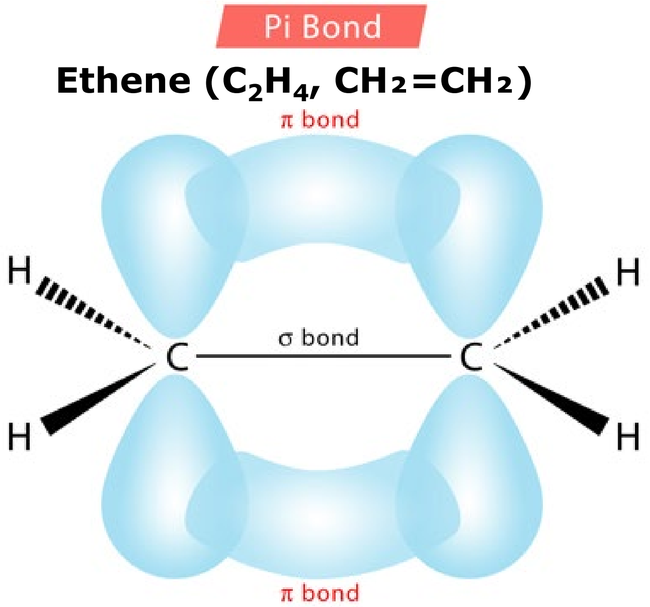
- Formed by end-to-end (head-on) overlap of orbitals (s, p, or hybrid)
- Maximum effective orbital overlap → very strong bond!
- Allows free rotation of bonded atoms



Covalent bond – π bond

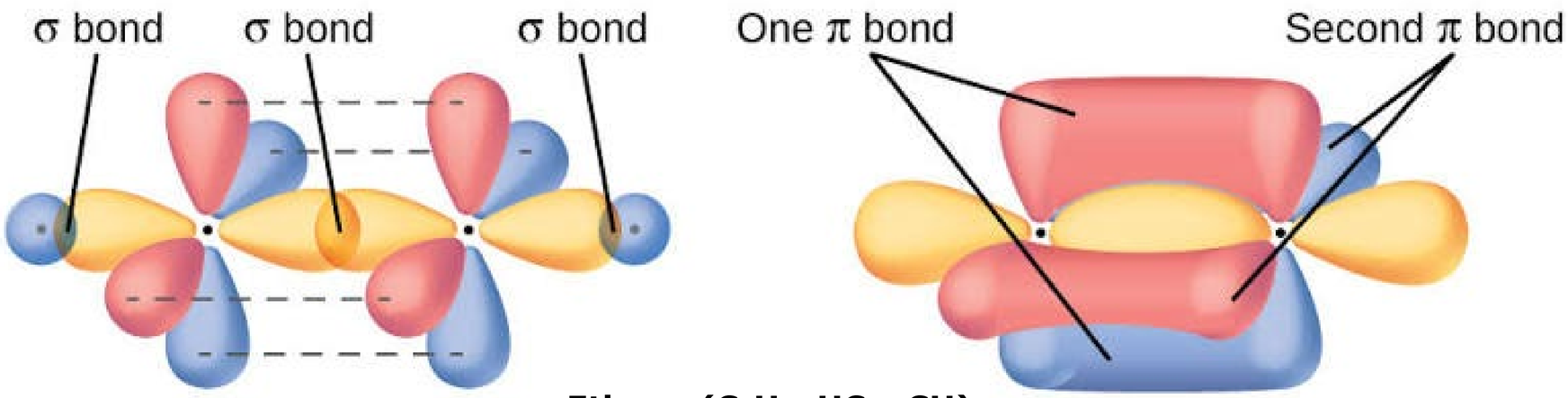
- Sideways overlap of p-orbitals
- It is a component of double and triple bonds → a π bond is not always a double bond, but a double bond always contains one π bond
- Weaker than σ bonds because sideways overlap is less direct!

  → BUT double bond > single bond, because double bond = σ + π.
- Restricts rotation → double and triple bonds are “stiff”
- More reactive than σ bond



Covalent bond – single vs. multiple

- **Single bond**: contains exactly one strong σ bond (1 pair of e^- shared)
- **Double bond**: contains one strong σ bond + 1 weaker π bond (2 pair of e^- shared)
- **Triple bond**: Contains one strong σ bond + 2 weaker π bond (3 pair of e^- shared)

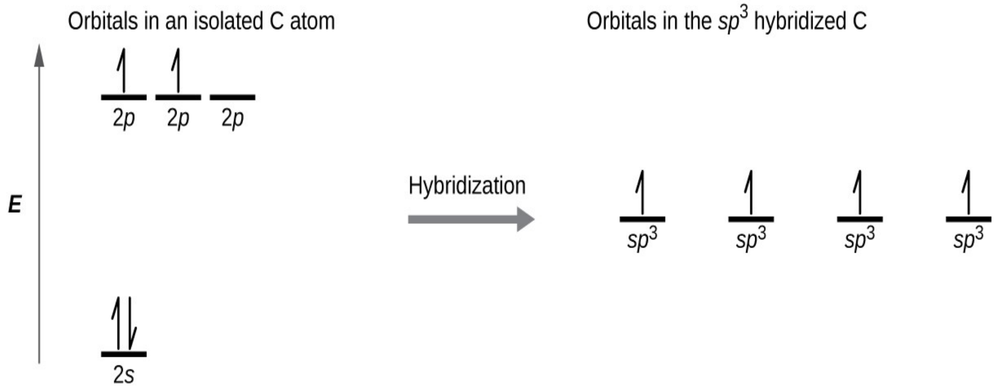


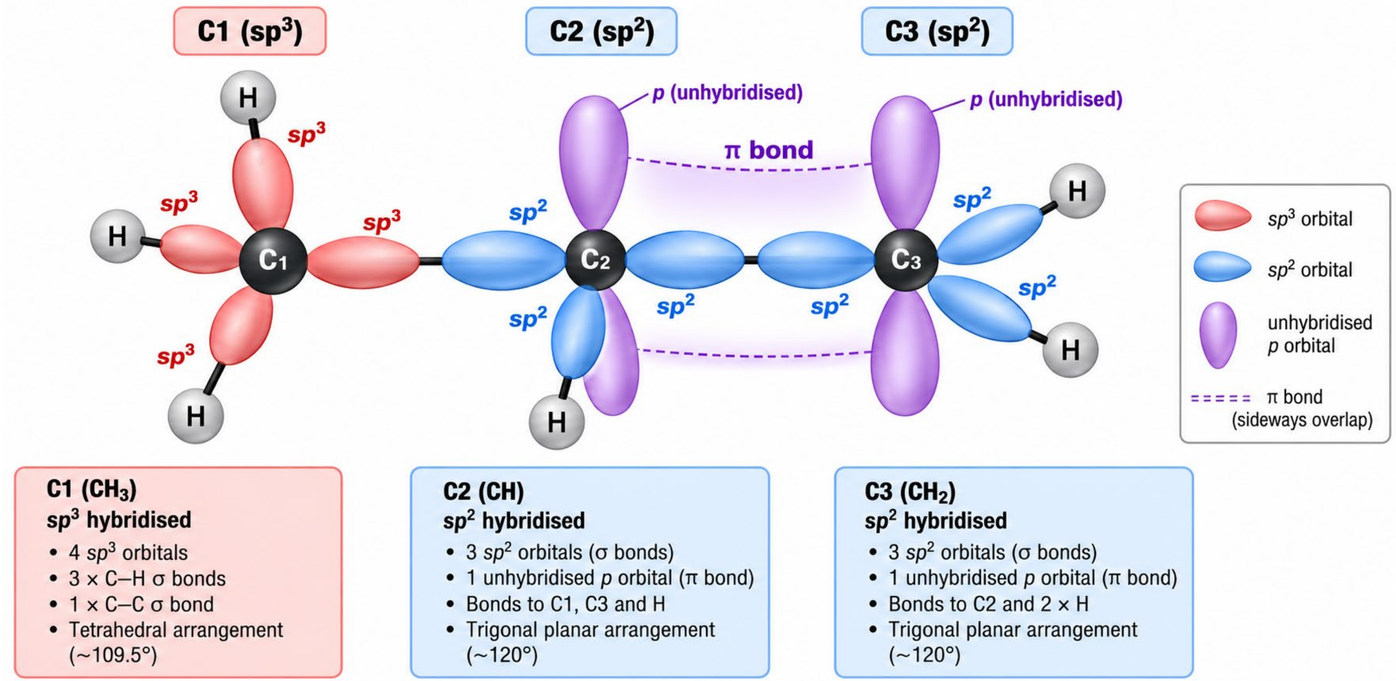
Ethyne (C_2H_2, HC ≡ CH)

- The more bonds, the stronger/stiffer the structure

Exercise

Determine the hybridization of $CH_3–CH=CH_2$

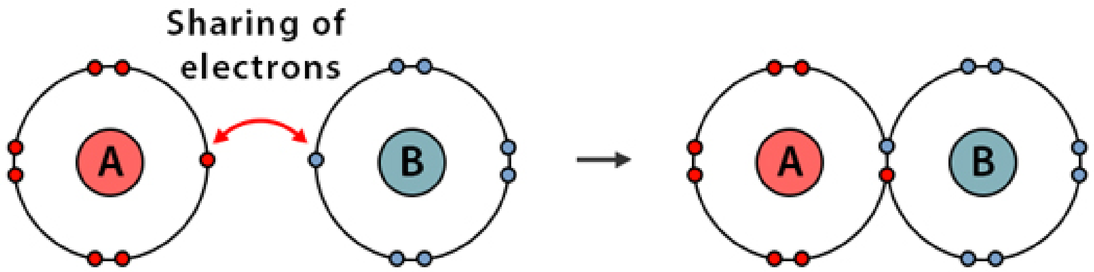




Covalent bond

When two atoms have similar electronegativity, the e^- are shared!

→**<u>Covalent</u>** bond → neither strongest/weakest → can be single, double, triple



Covalent bonds lead to:

- Low bond energies = low melting points (e.g. methane, CH4 = -180°C!)
- Generally poor heat and electrical conductivity

Polarity

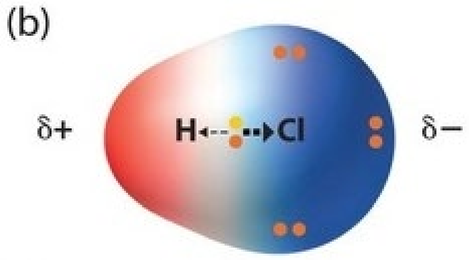
Depending on electronegativity, orbitals overlap can be symmetrical or asymmetrical



- If electronegativity is similar (difference < 1.0), the covalent bond is **non-polar**.
- Electron density is distributed equally

Polarity

Depending on electronegativity, orbitals overlap can be symmetrical or asymmetrical

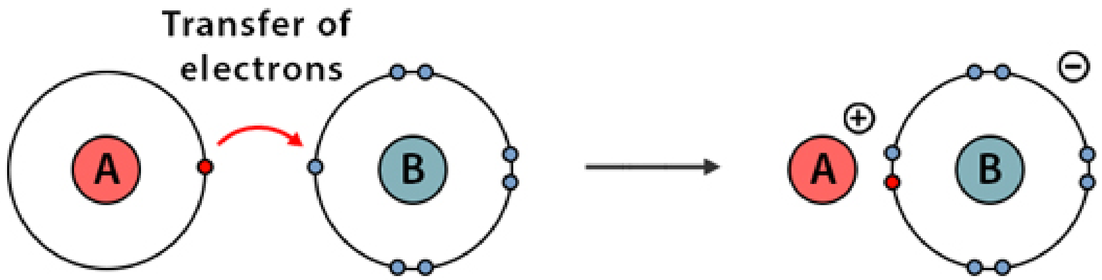


- If electronegativity is between 1.0 and 1.6, the covalent bond is **polar**.
- Electron density is distributed non-equally
- Creates a **dipole moment** → measures the separation of charge. Measured in **Debye units** → molecules with dipoles can interact through dipole – dipole interactions
- Usually polar molecules → **hydrophilic**, or water-liking!

Ionic bond

When the electronegativity difference is > 1.6, the e^- are donated/received!

→**ionic** bond → one of the strongest type of bond! Very **polar**

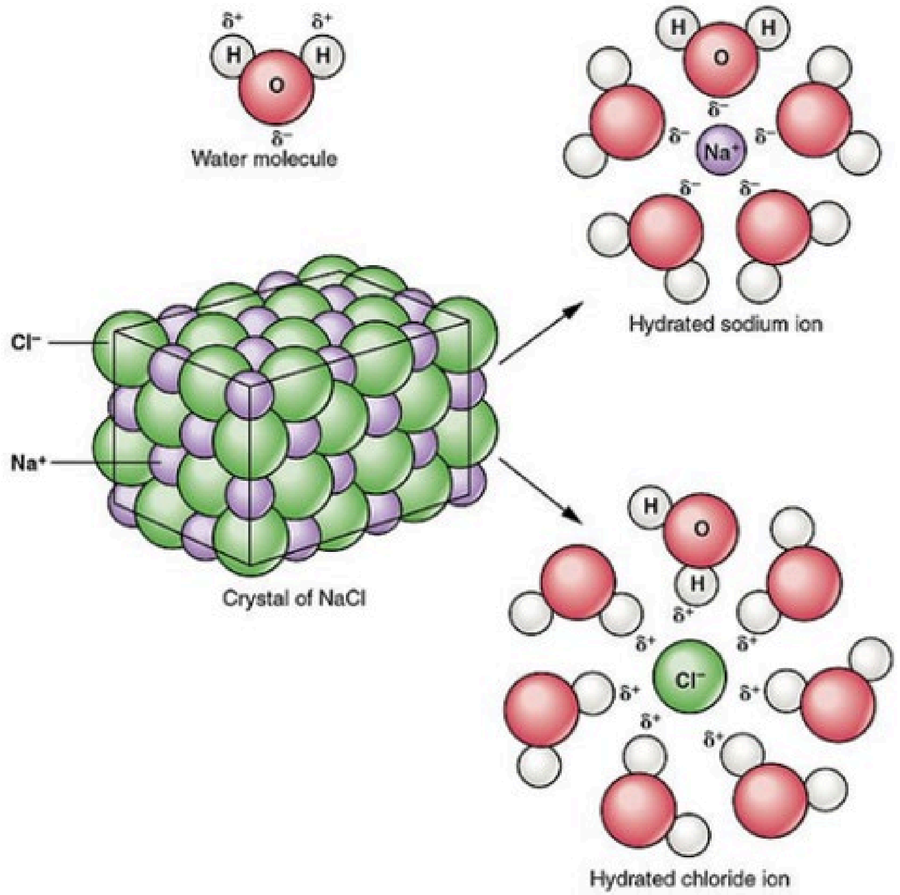


Ionic bonds lead to:

- High bond energies = higher melting points (e.g. NaCl = 800°C!)
- Solubility in water

Ionic bond

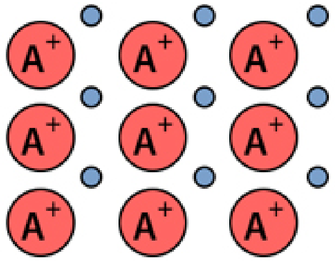
- Ionic compounds are **solvated** → structure is lost when dissolved → good electrical conductors in solution, but not when solid



Metallic bond

Special type → a “sea” of free e^- among a structure of positively charged ions (cations)

Typical of e^- deficient atoms (transition metals) → can’t reach octet, few valence e^-



Metallic bonds lead to:

- High ductility and strength (because e^- can “slide”)
- High electrical and thermal conductivity
- High melting points (because of strong electrostatic attractions)

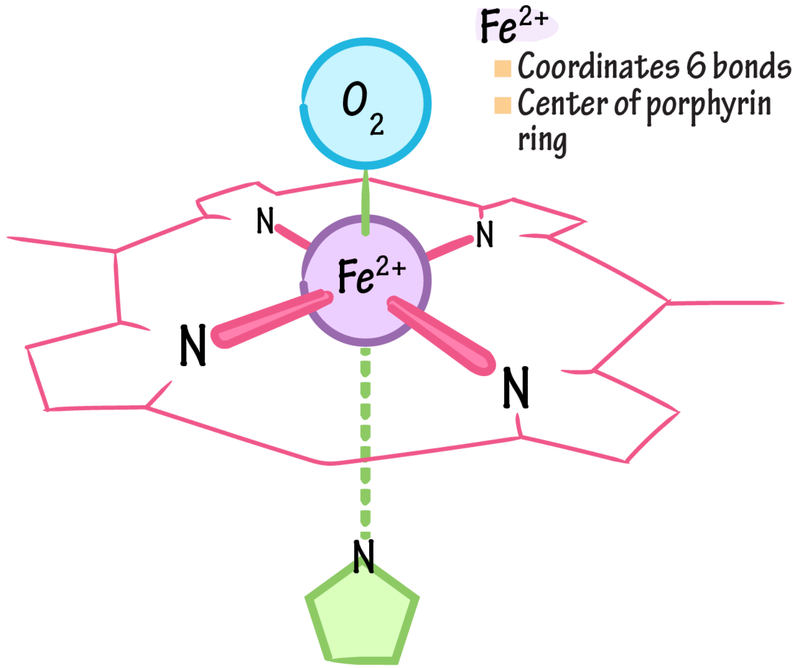
Coordination complexes

Chemical compounds made of:

1. central atom or ion (**coordination center**, usually transition metal)
2. surrounding array of bound molecules or ions (**ligands**)

In principle both species can exist on their own.

Both of the e^- shared come from the ligand!



Break
10:05 – 10:25

Types of bonds (intermolecular)

Hydrogen bond

”Hybrid” covalent and electrostatic. Can be both inter and intramolecular!

H acts as a “bridge” between an atom it’s covalently bound to, and another electronegative atom

Often happens with N, O and F



Hydrogen bonds leads to:

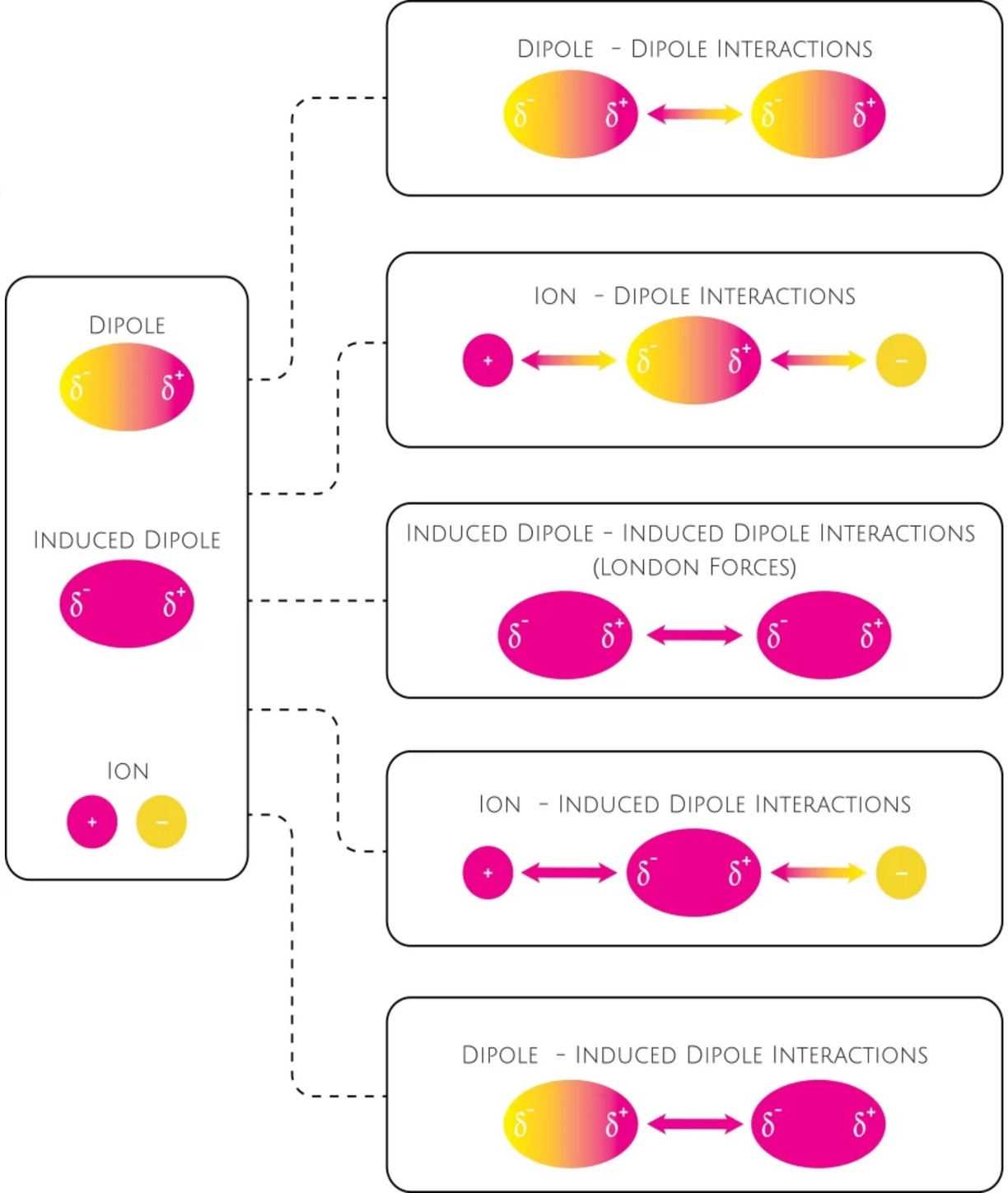
- High viscosities, high surface tension
- Higher melting/boiling points
- Solubility in polar solvents

Van Der Waals forces

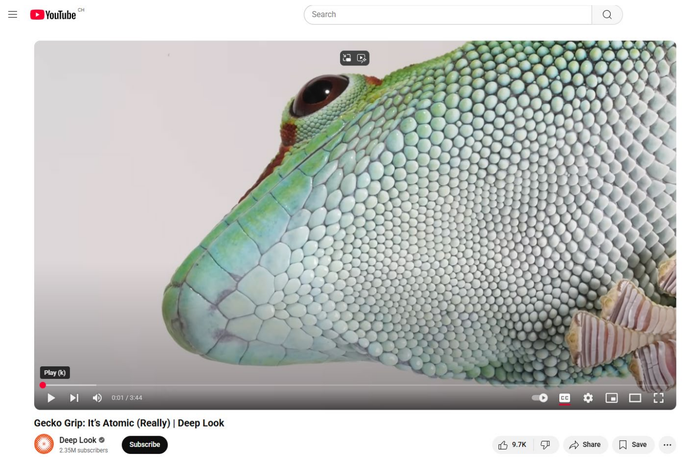
- Family of temporary, distance-dependent electrostatic interactions. Vanish at longer distances
- Generally considered rather weak.
- Caused by temporary imbalances in e^- that create temporary or permanent dipoles.
- Compounds that are more polarizable (more e^-) give rise to stronger VdW

VdW forces lead to:

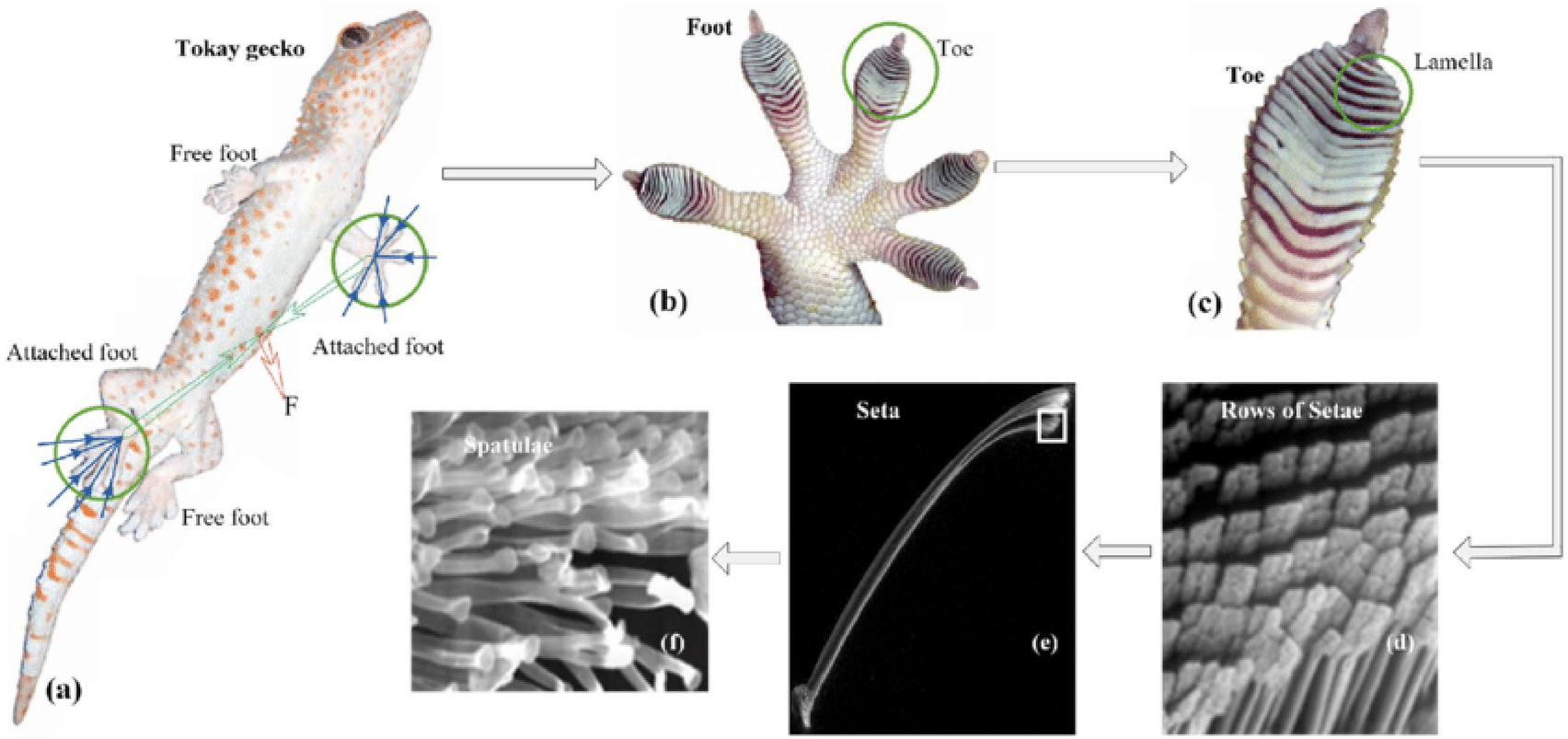
- Generally low melting points



Van Der Waals – The Gecko case study



https://youtu.be/r5cqJTU5D5I?si=CkfZc0rsO3neqcMH



DOI: 10.1007/s42235-021-00088-7

Exercise



Discussion + Q&A





Outlook for Next Week:

- Recap bonds + open doubts
- Lewis structures
- Exercises